Demystifying enhancer-gene linkage using population-scale single-cell multiomics

Tengfei Cui[1], Yang Ni[2*], Yuchao Jiang[1*]

1 Department of Statistics, Texas A&M University, College Station, Texas, United States of America

2 Department of Statistics and Data Sciences, The University of Texas at Austin, Austin, Texas, United States of America

* yang.ni@austin.utexas.edu; yuchaojiang@tamu.edu

Abstract

The epigenetic regulation of gene expression is highly cell-type and context specific. Recent advances in single-cell RNA and ATAC multiomic technologies provide unprecedented opportunities to map enhancers to their target genes. Existing methods, however, largely rely on enhancer-gene correlation either across individual pseudobulks or across heterogeneous cell types. We develop a Mendelian randomization framework leveraging population-scale single-cell multiomics to directly identify causal and pleiotropic enhancer-gene relationships, one cell type at a time. We apply our method to single-cell RNA and ATAC data of 428 individuals from the Chinese Immune Multiomics Atlas and show that the resulting enhancer-gene regulatory landscape differs substantially from those identified by existing approaches. We orthogonally validate our findings using promoter capture Hi-C and H3K27ac HiChIP data and demonstrate significant enrichment of both transcription factor motifs and blood-trait heritability among the identified enhancer-gene pairs.

Author Summary

Introduction

Results

Study overview and comparison of peak–gene linking methods

We applied MR methods to data from the Chinese Immune Multi-Omics Atlas (CIMA) [1], a multi-omics analysis of 428 healthy Chinese adults that combined scRNA-seq and scATAC-seq of circulating immune cells with whole-genome sequencing (WGS) of plasma cell-free DNA. CIMA applied TensorQTL [CITE: TensorQTL] to identify cell-type-specific *cis*-xQTLs within ±1 Mb of gene transcription start sites (TSSs) or chromatin peak midpoints. We used the cell-type-specific caQTL and eQTL summary statistics to estimate the effect of peak accessibility on gene expression in naive B cells (Bn cells), naive $CD4^+$ T cells (CD4 T cells), and naive $CD8^+$ T cells (CD8 T cells). MR estimates causal effects under the instrumental-variable assumptions, whereas correlation-based methods estimate associations. We used three established MR methods [2–5], which we call causal IVW, causal Egger and causal SMR, and an equal-weight Cauchy combination of these three tests (causal joint) [6]. We compared them with two correlation-based methods: Spearman correlation across individual-level pseudobulk profiles of CIMA individuals (marginal pseudobulk; 335 individuals for Bn cells, 372 for CD4 T cells, and 346 for CD8 T cells) and Spearman correlation across 80 WNN metacells of an external PBMC multiome dataset (marginal multiomics) (Fig 1A). We refer to the four MR-based tests (causal IVW, causal Egger, causal SMR and causal joint) collectively as the causal methods, and to the two correlation-based methods as the marginal methods. Significance was defined by the BH-adjusted P value computed within each gene, separately for each method ($P_{\mathrm{adj}} < 0.1$), and an MR-only link is a pair that is significant for at least one of causal IVW, causal Egger and causal SMR but for neither marginal method (see Methods).

All six methods were applied to the same set of candidate peak–gene pairs (40,087 for Bn cells, 85,527 for CD4 T cells, and 50,931 for CD8 T cells; S3 Table) for a fair comparison. Marginal multiomics found 37,181 links ($P_{\mathrm{adj}} < 0.1$) in CD4 T cells, many more than the other methods. Causal methods (causal IVW, causal Egger, and causal SMR) generally found fewer links than the two marginal methods (marginal multiomics and marginal pseudobulk). The sets of links overlapped between methods (Fig 1B). The results for Bn cells and CD8 T cells were similar (S3 Fig).

Even if some peak-gene pairs are significant in both marginal and causal methods, the directions might be different. We extracted all 1,259 links in CD4 T cells that were significant for all six methods and compared their effect directions. The results of causal IVW, causal Egger, and causal SMR were highly concordant

(1,258 of 1,259 links with the same sign). For 73% of these links, causal IVW had the same sign as both marginal methods, whereas for 24.1% the sign of causal IVW differed from that of marginal multiomics only. Marginal multiomics was more likely to be discordant with causal IVW than marginal pseudobulk (Fig 1C, S3 Fig).

Concordance and discordance between MR estimates and marginal correlations

Mendelian randomization estimates the causal effect of peak accessibility on gene expression, while marginal multiomics and marginal pseudobulk calculate the correlation between peaks and genes across metacells (spanning cell types) and across individuals, respectively. The causal effects and marginal associations agree on many peak-gene pairs, which is reasonable. But discordance still exsits. We select four representative case to illustrate this.

For the gene *SIDT2* and the peak chr11:117185715-117186216 in Bn cells, the MR results (causal SMR, causal IVW, causal Egger, and causal joint) and two marginal correlations (marginal multiomics and marginal pseudobulk) were all positive and significant ($P_{\text{adj}} < 0.1$) (Fig 2A). Causal IVW gave the smallest P value, and the two lines were closely aligned implying no clear pleiotropy in the data (Fig 2A).

TODO: R2-P2: delete the "no clear pleiotropy" clause, or compute the Egger intercept (author to decide).

However, the concordance of significance did not always exist among methods. For second examples (gene *RPL11*, peak chr1:23691494-23691995) in CD4 T cells, while the MR estimates and the correlation across metacells are significant, the correlation across individuals was not significant anymore (Fig 2B). Therefore, the causal effects were not always mirrored by a marginal association within the same cell type.

The causal effects and the marginal associations might have even different sign. For the gene *ERAP2* and the peak chr5:96885894-96886395 in CD4 T cells, all of causal IVW, causal Egger, and causal SMR detect negative and significant effects, consistent with the negative correlation across individuals ($\rho = -0.641$) (Fig 2C). However, the correlation across cell types was surprisingly positive ($\rho = 0.375$, $P = 6.1 \times 10^{-4}$), indicating the possible directional discrepancy between causal effects and marginal association.

Some links were detected only by MR methods. For example, all MR methods detected a significantly positive effect of the peak chr7:99376204-99376705 on *ARPC1B* in CD4 T cells, whereas neither marginal correlation test provided evidence of association (Fig 2D).

Together, these examples show the consistency and possible difference between marginal correlation and causal effects. Because marginal multiomics and marginal pseudobulk only calculate the marginal association, which might be influenced by potential unobserved counfunding factors, the calculated correlation may be

different or even of different signs from the causal effects. This demonstrates that when studying the regulation mechanism between peak accessibility and gene expression, we should not only study the marginal association, but also find the underlying causal effects.

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Functional characterization and regulatory network support for peak–gene links

TODO: Fig 3A (Miami plot, CD4 T cells) and Fig 3D (HiChIP overlap of MR-only links; S5 Fig) are not yet cited in the text; S7.1 Fig–S7.3 Fig (genome-wide Manhattan plots) is not yet cited either (author to write).

To characterize the biological functions of MR-only links (see Methods), we performed GO enrichment analysis using DAVID [CITE: DAVID] on their genes. We pooled the MR-only links from the three cell types, giving 4,509 MR-only genes (4,031 with a GO Biological Process annotation). DNA-templated transcription was the most significant biological process term, including 689 genes, with a fold enrichment of 1.36 and a DAVID-reported FDR of 5.18×10^{-18} (Fig 3B). The molecular function enrichment showed protein binding (fold enrichment 1.16, DAVID FDR 2.81×10^{-85}), RNA binding (fold enrichment 1.41, DAVID FDR 3.58×10^{-15}), and zinc ion binding (fold enrichment 1.33, DAVID FDR 3.58×10^{-15}) as the most significant terms (Fig 3C). Genes related with the RNA-binding enrichment, such as *SRSF7* amd *DDX5*, were known to have establilshed role in pre-mRNA splicing and RNA processing [CITE: SRSF7 and DDX5 in pre-mRNA splicing and RNA processing]. Genes included in the zinc ion binding term included the zinc-finger proteins *ZNF143* and *IKZF3* [CITE: ZNF143 and IKZF3 as zinc-finger regulators].

By integrating scRNA-seq and scATAC-seq data, CIMA applied SCENIC+ (version 1.0.1) [CITE: SCENIC+] to construct cell-type specific enhancer-driven regulons (eRegulons) and further selected a subset of high-quality eRegulons [1]. Each eRegulon is a triple of (TF, chromatin peak, gene), which can reasonably serve as validation data. We extracted unique region-gene pairs from the CIMA eRegulons and calculated the odds ratio for overlap with the links ($P_{\text{adj}} < 0.05$) identified by each method (Fig 3E for CD4 T cells, S8 Fig for Bn cells and CD8 T cells; see Methods). Although marginal multiomics and marginal pseudobulk tended to find more links (Fig 1B, S3 Fig), causal methods had higher odds ratios than the marginal methods in most comparisons across cell types. Although overlap with the CIMA regulatory network is not a ground truth for causal regulatory effects, it could partially support the reliability of MR estimates.

We also tested whether the significant peaks of each method were enriched for peaks overlapping six external regulatory references: EnhancerAtlas, FANTOM5, 4DGenome, GTEx whole-blood eQTLs, H3K27ac HiChIP

and promoter capture Hi-C (PCHi-C) (see Methods; S1 Table and S2 Table). The significant peaks of all six methods were enriched for every reference in every cell type (pooled hypergeometric $P < 10^{-16}$; fold enrichment 1.68–6.18). Causal SMR had the largest pooled fold enrichment in 17 of the 18 combinations of cell type and reference; the exception was PCHi-C in CD4 T cells, where causal IVW was largest (2.61 versus 2.55 for causal SMR). Marginal multiomics had the smallest fold enrichment in all 18 combinations. For example, in Bn cells the fold enrichment of causal SMR and marginal multiomics was 6.18 and 3.56 for FANTOM5, and 4.46 and 3.08 for HiChIP. The fold enrichments are descriptive and have no confidence intervals; the hypergeometric P values depend on the number of significant peaks and were often smaller for the marginal methods, which had more significant peaks but lower fold enrichment.

Disease heritability enrichment and GWAS overlap of MR-supported peak–gene links

To explore the disease relevance of MR-only links, we assessed heritability enrichment using stratified LD score regression (S-LDSC) [7]. We pooled the peaks of MR-only links (see Methods) from the three cell types and obtained 3,777 unique peaks after liftOver to hg19 (MR-only). We constructed an annotation from these 3,777 peaks and fitted the S-LDSC model together with the 97 baselineLD v2.2 annotations. For comparison, all candidate peaks (All peaks) were analyzed with the same S-LDSC pipeline (Fig 4A). We select the seven GWAS triated studied in OneK1K paper (OneK1K reference).

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Enrichment scores were defined as the ratio of the proportion of heritability explained to the proportion of variants. MR-only peaks had higher enrichment point estimates than all candidate peaks across the seven immune-mediated diseases (Fig 4A). Nominal evidence of enrichment at $P < 0.1$ was observed for Crohn's disease, inflammatory bowel disease, multiple sclerosis, and type 1 diabetes (Fig 4A).

We next showed individual MR-only links whose peak regions contained genome-wide significant variants. These were strong MR-only links, with $P_{\text{adj}} < 0.001$ for at least one causal method and $P_{\text{adj}} \geq 0.05$ for both marginal methods (see Methods). For example, an MR-only peak linked to the gene *ERBB3* in CD4 T cells contained rs705704, a variant significantly associated with type 1 diabetes (Fig 4B). This finding nominated *ERBB3* as a candidate target gene of the GWAS-overlapping regulatory region in CD4 T cells. Additional examples included *RGL2*-linked peaks overlapping type 1 diabetes and rheumatoid arthritis variants, a *POLR2E*-linked peak overlapping Crohn's disease variants, a *GNAS*-linked peak overlapping inflammatory bowel disease and Crohn's disease variants, and an *ERAP1*-linked peak in Bn cells overlapping a Crohn's disease variant (Fig 4B). Although we cannot directly say there were causal relationship between the genes

and GWAS diseases, these MR-supported peak-gene links provided a potential regulation mechenism ways for these diseases.

Motif enrichment highlights candidate regulators of MR-supported peak–gene links

To characterize the regulatory sequence features of the peaks of each method, we performed transcription factor motif enrichment analysis (see Methods). For each cell type and method, the peaks of links with $P_{\mathrm{adj}} < 0.1$ formed the foreground set, and the background was 50,000 candidate peaks matched for GC content and chromatin accessibility. We focused on MR-specific motifs, which were enriched for at least one of causal IVW, causal Egger and causal SMR but for neither marginal method; there were 47 such motifs (23 in Bn cells, 14 in CD4 T cells, and 10 in CD8 T cells). Table 1 shows the most significant MR-specific motifs for each cell type. All six motifs in Table 1 were enriched for causal Egger, and four of them (FOSL2::JUND, MNT, POU3F1 and TEF) were enriched only for causal Egger; causal Egger estimates depend on the allele orientation of the instruments (see Methods; S2 Fig).

In CD4 T cells, the MNT motif had the strongest enrichment (motif-level adjusted $P = 0.003$, fold enrichment 1.34 for causal Egger); loss of MNT has been reported in malignant $CD4^+$ T cells from patients with Sézary syndrome [8]. In Bn cells, ATF3 had the strongest enrichment, and causal Egger-selected peaks were also enriched for the FOSL2::JUND motif, and JunD, one of its two AP-1 components, is upregulated in B cells from patients with SLE [9]. Other selected motifs included CENPB in CD4 T cells, and POU3F1 and TEF in CD8 T cells (Table 1).

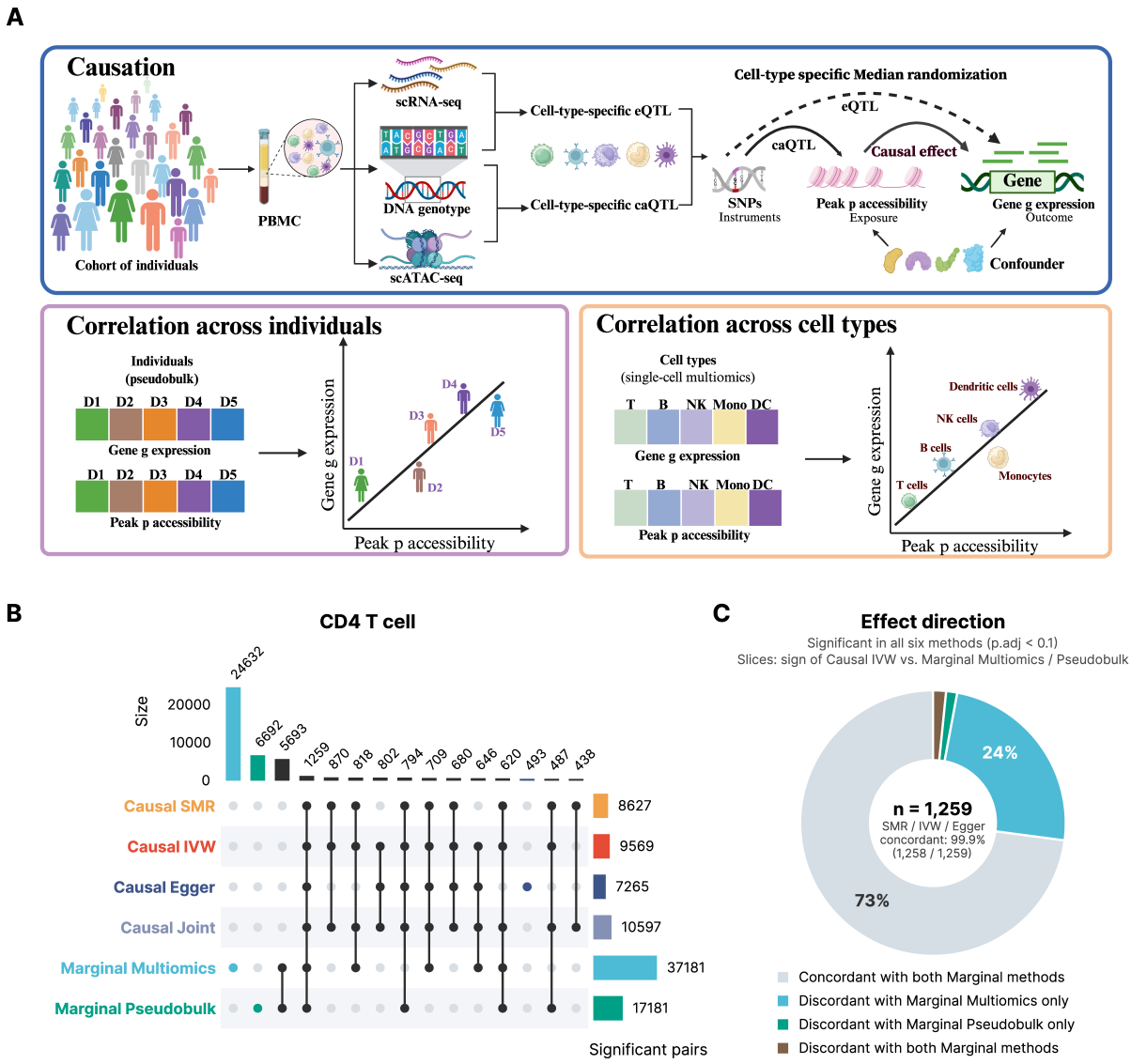


Fig 1. The complete analysis framework and the results of CD4 T cells. **A** MR methods (IVW, MR Egger, SMR) use cell-type specific caQTL/eQTL summary statistics; Marginal multiomics uses external PBMC metacells, and marginal pseudobulk correlations use CIMA individuals. Both two marginal correlation are measured by the Spearman correlation. **B** UpSet plot of significant links in CD4 T cells ($P_{\text{adj}} < 0.1$) **C** Effect directions among the 1,259 CD4 T cell links significant in all six methods. Slices compare the sign of the IVW estimate with the signs of the marginal correlations: concordant with both marginal methods (light gray), discordant with Marginal Multiomics only (blue), discordant with Marginal Pseudobulk only (green), and discordant with both (brown). Bn and CD8 T cell comparisons appear in S3 Fig.

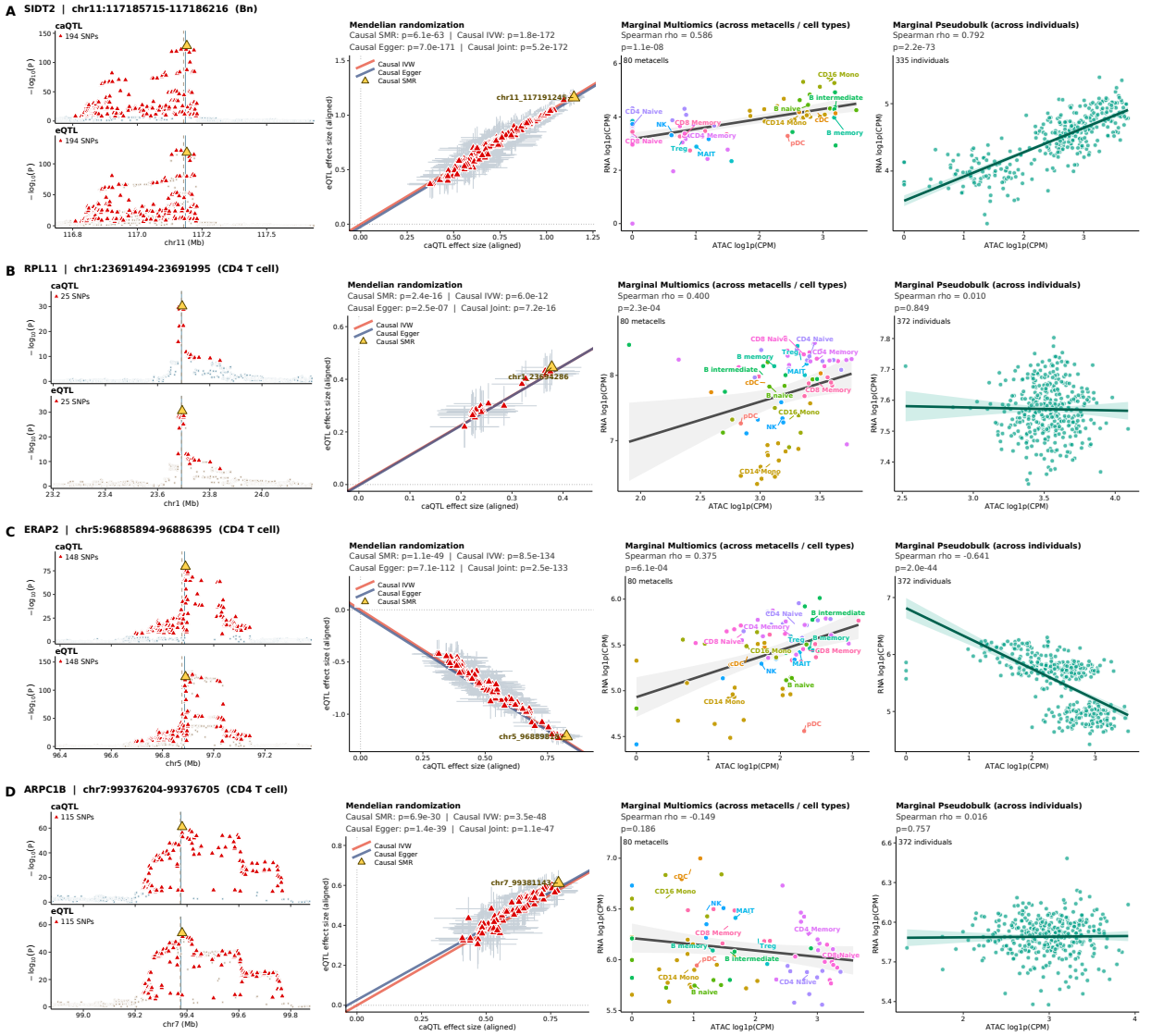


Fig 2. Four selected peak–gene links. Rows show four peak–gene pairs: **A,** *SIDT2* in Bn cells; **B,** *RPL11* in CD4 T cells; **C,** *ERAP2* in CD4 T cells; **D,** *ARPC1B* in CD4 T cells. **First column:** Regional caQTL (top) and eQTL (bottom) tracks. The horizontal axis indicates genomic position, and the vertical axis shows $-\log_{10}(p)$. Red triangles mark retained MR instruments, and yellow triangles identify the SNP selected for SMR. **Second column:** MR method results, red and blue lines show the IVW and MR-Egger fits, respectively. The yellow triangle highlights the leading SNP used in SMR. Report p-values correspond to SMR, IVW, MR-Egger, and the Cauchy combination ("Causal Joint"). **Third column:** Peak accessibility and gene expression across 80 WNN metacells from an external PBMC multiome dataset. Each point represents a metacell, colored by cell type **Fourth column:** Peak accessibility and gene expression across matched individuals. Each point represents an individual.

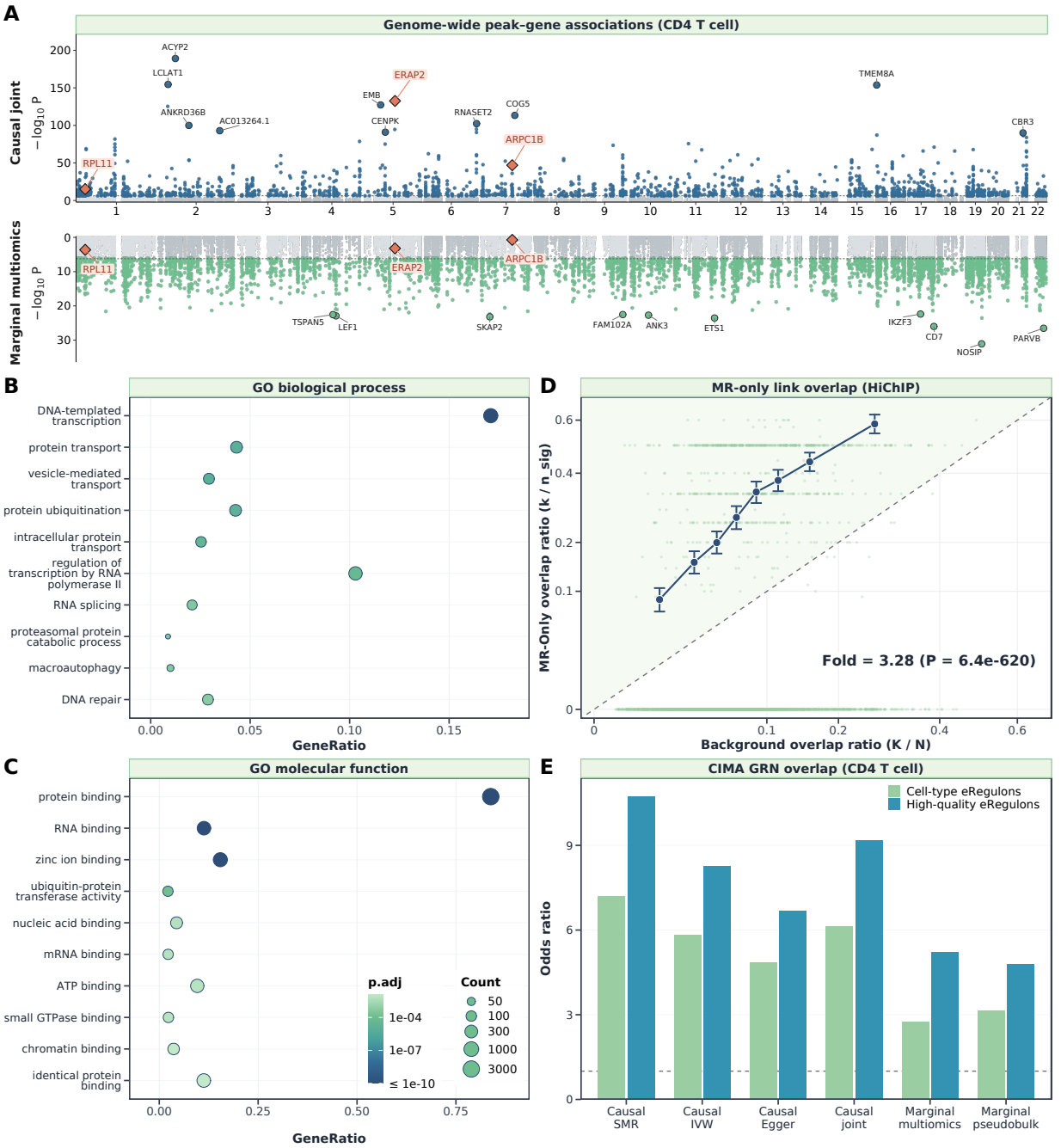


Fig 3. Genome-wide view, GO enrichment, regulatory-reference overlap and regulatory network overlap of peak–gene links. **A,** Miami plot of the 85,527 pairs tested in CD4 T cells (peak center, hg38): $-\log_{10} P$ of causal joint (top) and marginal multiomics (bottom, mirrored). Dashed lines, Bonferroni threshold (0.05/85,527); labels, the strongest pair of selected genes above it (at most one per 5 Mb); diamonds, the example pairs of Fig 2 (*RPL11*, *ERAP2*, *ARPC1B*). Caption continued on the next page.

Fig 3 (continued). B, GO Biological Process terms (4,031 annotated genes) and **C,** GO Molecular Function terms (4,185 annotated genes) for the genes of MR-only links pooled across the three cell types (top 10 terms by DAVID FDR; x-axis, GeneRatio; point size, term count; color, DAVID FDR, shown as p.adj, with values below 10^{-10} in the darkest color). **D,** Overlap of MR-only links with HiChIP interactions compared with all candidate peaks (panel E of S5 Fig, where the axes and groups are defined); the lower right gives the pooled fold enrichment and hypergeometric P value. **E,** Odds ratio for a link significant for each method in CD4 T cells (within-gene $P_{adj} < 0.05$, a stricter threshold than the $P_{adj} < 0.1$ used elsewhere) being present in the CIMA SCENIC+ regulatory network, compared with links not in the network (2×2 table; all tested pairs in the cell type as background). Light bars use eRegulons active in the cell type (cell-type eRegulons; 2,520 tested pairs), and dark bars the subset in the high-quality eRegulons of the lineage (high-quality eRegulons; 458 tested pairs). Fisher's exact test $P < 10^{-30}$ for every bar; the dashed line marks an odds ratio of 1. Bn cells and CD8 T cells are shown in S8 Fig.

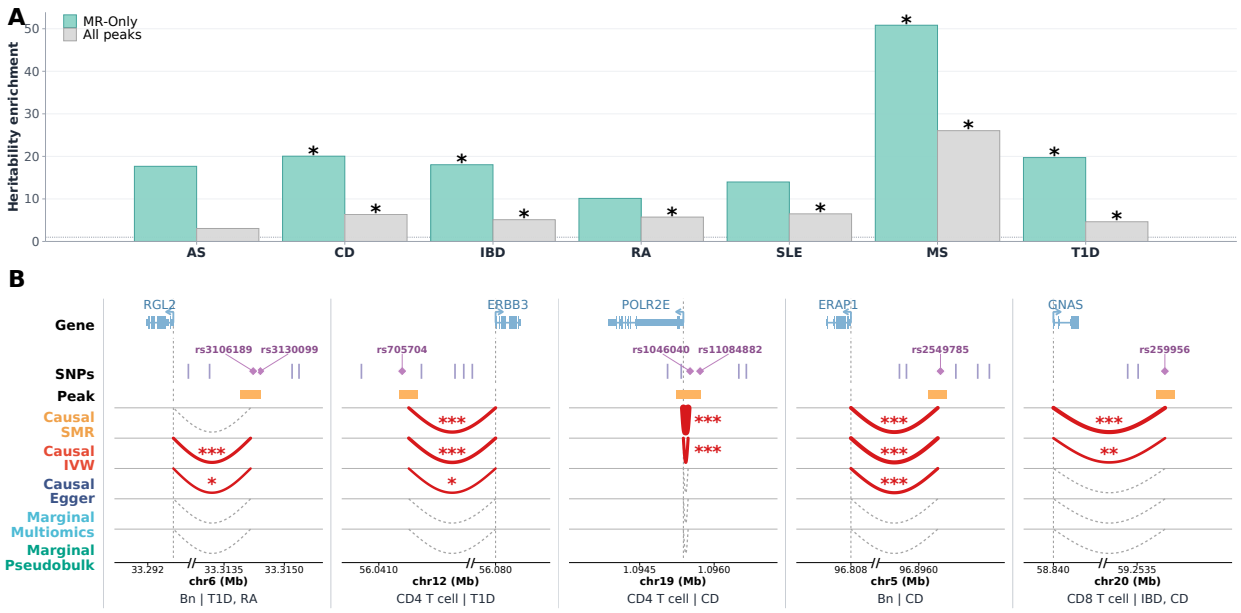
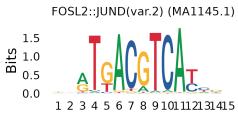
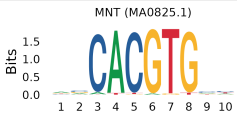
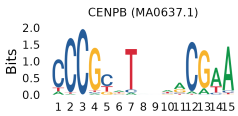
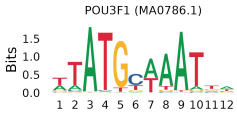
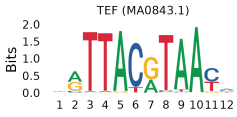


Fig 4. Disease-heritability enrichment and GWAS-overlapping regulatory links. **A** Heritability enrichment of the combined MR-only and all-candidate-peak S-LDSC annotations for each GWAS trait, shown on a common y axis; asterisks mark nominal enrichment $P < 0.1$. **B** Tracks show the gene, MR instruments (lavender ticks), GWAS variants within the peak (purple diamonds), the peak (orange bar), and one link per method (Causal SMR, IVW, and Egger; Marginal Multiomics and Pseudobulk). Significant links ($P_{adj} < 0.05$) are drawn in red, and line width increases with $-\log_{10} P_{adj}$; gray dashed links are nonsignificant. Link stars denote $P_{adj} < 0.05$ (*), 0.01 (**), and 0.001 (***).

Table 1. MR-specific transcription factor motifs in peaks supported by Mendelian randomization.

Motif logo	p.adj						Brief interpretation
	Causal IVW	Causal Egger	Causal SMR	Causal Joint	Marginal Multi-omics	Marginal Pseudo-bulk	
Bn							
	**0.041**	**1.5e−4**	0.157	0.059	0.072	0.056	Integrating gene expression with H3K4me3 changes in T cells, B cells and monocytes from patients with SLE identified ATF3 as a pivotal transcription factor in the disease process [10].
	0.057	**0.001**	0.137	0.056	0.088	0.059	JunD, one partner of this AP-1 heterodimer motif, is upregulated in B cells from patients with SLE, where it regulates B-cell proliferation, autophagy and immune function [9].
CD4 T cell							
	0.176	**0.003**	0.051	**0.032**	0.077	0.113	Loss of the MYC antagonists MNT and MXI1 was found in malignant CD4$^+$ T cells from 40–55% of patients with Sézary syndrome [8].
	**0.035**	**0.004**	0.172	**0.033**	0.329	0.156	CENP-B is a major anti-centromere autoantigen; antibodies to recombinant CENP-B were detected in 60% of patients with primary biliary cholangitis [11].
CD8 T cell							
	0.415	**0.006**	0.591	0.140	0.499	0.232	POU3F1 is one of the top five transcription factors predicted from disease-associated chromatin accessibility and gene expression in CD4$^+$ T cells of SLE patients [12].
	0.112	**0.010**	0.067	0.124	0.117	0.087	TEF was identified as a novel TCF3 fusion partner (TCF3::TEF) in pediatric B-lymphoblastic leukemia/lymphoma [13].

Motif enrichment was tested for peaks selected by each method (within-gene BH-adjusted $P_{\text{adj}} < 0.1$) against a background matched on GC content and accessibility (Signac, JASPAR2020 vertebrate CORE motifs). MR-specific motifs have p.adj < 0.05 and fold enrichment > 1 for at least one of causal IVW, causal Egger and causal SMR, with no significant enrichment (p.adj ≥ 0.05) for either marginal method; for each cell type, the two with the smallest p.adj among the causal methods and a clear literature-based interpretation are shown. Bold, p.adj < 0.05 with fold enrichment > 1. p.adj is the adjusted motif-enrichment p-value reported by Signac for each method. Each interpretation is based on the single paper cited in the same row. Causal joint, Cauchy combination of causal SMR, causal IVW and causal Egger; marginal multiomics, Spearman correlation across WNN metacells; marginal pseudobulk, Spearman correlation across individual pseudobulks.

Discussion

There is one limit of our study. MR usually study Two sample data, where the variants-exposure and variants-outcome comes from two independent study. The two sample setting is important to get the causal effects. However, in our study, caQTL and eQTL data come from the same population. This might introduce some bias towards the true causal effects. We did sensitivity analysis by selecting SNPs under the null hypothesis and checked the correlations of residuals (S1 Fig). Both of global and peak-gene pair-specific correlation support that there is little correlations between the two marginal associations in our study. The boxplots show the p values and correlation estimates for different number of SNPs, where more SNPs in one peak-gene pairs indicate the clearer independence of residuals. The correlation between two residuals might be negligible. In the further study, we can also borrow eQTL data and caQTL from two separately study to eliminate the poential bias.

Another issue we want to discuss is the orientation of the genetics variants. Association of genetics variants can be quoted with respecrt to either the major or the minor allele. Causal IVW, which lack of intecept in the regression, provide exactly the same estimates using different of orientation of alleles (S2 Fig). However, causal Egger results would be different. we follow the common setting of MR Egger [CITE: MR-Egger allele orientation convention], and orientate the genetic variants so that the association with the risk factor all have the positive sign. This means that the directional pleitropy is define with respect to the risk factor-increasing allele. (S2 Fig)

Materials and Methods

Data sources

We used data from the Chinese Immune Multi-Omics Atlas (CIMA; OMIX009786) [1], which provides cell-type-specific caQTL and eQTL summary statistics together with single-cell RNA and ATAC count matrices. For the marginal multiomics method, we used an external single-cell multiome dataset of PBMCs (10x Genomics, granulocyte-sorted PBMCs, 10k cells [confirm]) [CITE: external PBMC multiome dataset].

TODO: Confirm the source of the external PBMC multiome dataset (notes/2_BOARD.md section 3).

CIMA provides hierarchical cell-type annotations. We focused on three major naive lymphocyte populations: Bn_TCL1A (naive B cells; marker gene *TCL1A*), CD4_Tn_CCR7 (naive $CD4^+$ T cells; marker gene *CCR7*), and CD8_Tn_CCR7 (naive $CD8^+$ T cells; marker gene *CCR7*), hereafter referred to as Bn cells, CD4 T cells, and CD8 T cells.

Study design and terminology

For each candidate peak–gene pair, we used caQTL variants as instruments for chromatin accessibility at the peak (the exposure) and eQTL summary statistics for expression of the gene (the outcome), and estimated the effect of accessibility on expression by Mendelian randomization (MR) within each cell type. We compared these MR estimates with two correlation-based measures of association between the same peak and gene. The MR estimators we used are established methods [2–6]; we retain their original names and describe our implementation in the following subsections.

We refer to the four MR-based tests (causal IVW, causal Egger, causal SMR, and causal joint) collectively as the causal methods, and to the two correlation-based methods as the marginal methods. Causal IVW and causal Egger are the generalized least squares (GLS) versions of the inverse-variance weighted (IVW) and MR-Egger estimators for correlated instruments, causal SMR is summary-data-based MR, and causal joint combines the P values of causal IVW, causal Egger, and causal SMR by the Cauchy combination test. The marginal methods are Spearman correlations between peak accessibility and gene expression, computed across WNN metacells of an external PBMC dataset (marginal multiomics) or across CIMA individuals within a cell type (marginal pseudobulk). The two labels reflect what each group estimates: MR estimates the causal effect of peak accessibility on gene expression under the instrumental-variable assumptions, whereas the marginal methods estimate associations. P values were adjusted by the Benjamini–Hochberg (BH) procedure within each gene, separately for each method; we denote this BH-adjusted P value by P_{adj} and, unless stated otherwise, call a result significant when $P_{\mathrm{adj}} < 0.1$. Each tested peak–gene combination is a peak–gene pair, and a pair that is significant for a given method is a link of that method. An MR-only link is a pair with $P_{\mathrm{adj}} < 0.1$ for at least one of causal IVW, causal Egger, and causal SMR and with $P_{\mathrm{adj}} \geq 0.1$ for both marginal methods; causal joint is not used in this definition. MR-only links are thus supported by MR but not detected by either correlation-based method, and we used them in the gene ontology and heritability enrichment analyses. Details of each method and of the filtering are given in the following subsections.

Data Preprocessing

Candidate peak–gene pairs. CIMA mapped eQTLs for each gene using all SNPs within 1 Mb of its transcription start site (TSS), and caQTLs for each chromatin accessibility peak using all SNPs within 1 Mb of the peak midpoint [1]. We considered all genes that had CIMA eQTL results and were present in the pseudobulk RNA data, together with all peaks in the CIMA ATAC data. The TSS of each gene was recovered from the eQTL results as the variant position minus the reported distance to the gene start. A peak–gene pair was retained as a candidate if the distance between the TSS and the peak center was at most 1 Mb.

Instrument selection. To satisfy the relevance assumption, we retained only caQTL variants with nominal $P < 10^{-8}$ in each cell type. We then performed LD clumping separately for each peak–gene pair with PLINK (v1.90b7.2) through `ieugwasr::ld_clump_local` (ieugwasr 1.1.0), using an r^2 threshold of 0.99, a window of 20,000 kb that covers the entire *cis* region, index variants ranked by caQTL P value, and `clump_p` = 1. The purpose of clumping was not to obtain independent SNPs but to avoid near-perfect collinearity among highly correlated instruments; the remaining correlation was modeled by GLS (Causal IVW: GLS-IVW with correlated instruments). We kept pairs for which the caQTL and eQTL data shared at least three variants and for which both the gene and the peak were present in the pseudobulk and WNN data (Marginal pseudobulk correlation and Marginal multiomics: WNN metacell correlation). At least three instruments were required to remain after each subsequent filtering step.

Allele harmonization and duplicate instruments. In both the caQTL and eQTL summary statistics, the alternative (ALT) allele was the effect allele. CIMA had harmonized the two sets of summary statistics, and we verified that their effect and non-effect alleles agreed; variants with discordant alleles would have been removed. Some SNPs had nearly identical caQTL and eQTL associations. A SNP was removed if both its caQTL and eQTL slopes differed by at most 10^{-3} from those of a previously retained SNP. Because SNPs were processed in variant-ID order, the first SNP of each such set was kept.

LD matrix. The signed LD correlation matrix $\boldsymbol{R}$ among the retained instruments was computed with PLINK (v1.90b7.2) using `-r square -keep-allele-order -write-snplist`, so that the correlations refer to the same effect alleles as the summary statistics. Variants absent from the reference panel, as well as rows and columns with undefined (NaN) correlations, were removed. The covariance matrix used in GLS was $\boldsymbol{\Sigma} = \mathrm{diag}(s_Y)\, \boldsymbol{R}\, \mathrm{diag}(s_Y)$ (Causal IVW: GLS-IVW with correlated instruments). Instruments were then oriented so that all caQTL effects were positive, and $\boldsymbol{R}$ was reoriented accordingly (Allele orientation). For both clumping and LD estimation, the reference panel was the East Asian (EAS) subset ($n = 585$) of the 1000 Genomes Project GRCh38 release (3,202 individuals) [CITE: 1000 Genomes GRCh38 high-coverage release]; chromosome X was coded as 23.

TODO: Confirm that the two EAS panel files used for clumping and for the LD matrix contain the same variant set (A1-09).

Mendelian randomization

Mendelian randomization (MR) uses genetic variants as instrumental variables (IVs) to estimate the causal effect of an exposure on an outcome. A SNP is a valid instrument if it satisfies three assumptions:

1. **Relevance:** the SNP is associated with the exposure.

2. **Independence:** the SNP is independent of any confounder of the exposure–outcome relationship.

3. **Exclusion restriction:** the SNP affects the outcome only through the exposure.

MR can be performed with summary statistics. For SNP j ($j = 1, \ldots, J$), these comprise the estimated SNP–exposure association $\hat{\gamma}_j$ with standard error s_{Xj} and the estimated SNP–outcome association $\hat{\Gamma}_j$ with standard error s_{Yj}, both aligned to the same effect allele. Under a linear model with a common causal effect β, the population-level associations satisfy

$$\Gamma_j = \beta\gamma_j + \alpha_j, \tag{1}$$

where $\beta\gamma_j$ is the component mediated by the exposure and α_j is the component not mediated by it. A SNP with $\alpha_j \neq 0$ is horizontally pleiotropic and violates the exclusion restriction. For a valid instrument, $\alpha_j = 0$, and the ratio Γ_j/γ_j identifies β.

With multiple instruments, MR-Egger regresses the SNP–outcome associations on the SNP–exposure associations with an intercept:

$$\hat{\Gamma}_j = \alpha_0 + \beta\hat{\gamma}_j + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \phi s_{Yj}^2), \tag{2}$$

where the slope β is the causal effect and ϕ is a residual variance multiplier. Under the Instrument Strength Independent of Direct Effect (InSIDE) assumption, which requires the pleiotropic effects α_j to be uncorrelated with the SNP–exposure associations γ_j, the intercept α_0 can be interpreted as the average pleiotropic effect of the instruments. Constraining the intercept to zero ($\alpha_0 = 0$) gives the inverse-variance weighted (IVW) estimator, a weighted regression through the origin. IVW gives consistent estimates when all instruments are valid, and also when the average pleiotropic effect is zero (balanced pleiotropy) and InSIDE holds; MR-Egger additionally allows for directional pleiotropy under InSIDE [3, 4]. The IVW and MR-Egger estimators for uncorrelated variants and the conditions under which they are consistent are given in S1 Text.

In this study, the exposure was chromatin accessibility at an ATAC-seq peak and the outcome was expression of the paired gene. For each candidate peak–gene pair in each cell type, $\hat{\gamma}_j$ and s_{Xj} were taken from the CIMA caQTL summary statistics, and $\hat{\Gamma}_j$ and s_{Yj} from the CIMA eQTL summary statistics (Data Preprocessing). Because the instruments of a pair lie in the same *cis* region and are correlated through LD, we used the GLS versions of IVW and MR-Egger described below.

Causal IVW: GLS-IVW with correlated instruments

SNPs within *cis*-regulatory regions are often correlated because of linkage disequilibrium (LD). Clumping stringently enough to obtain independent instruments can leave too few variants per peak–gene pair and thus limit statistical power. Causal IVW therefore uses the GLS extension of IVW, which accounts for the LD that remains among instruments after clumping [2,4].

For a given peak–gene pair with J instruments, let $\hat{\boldsymbol{\gamma}} = (\hat{\gamma}_1, \ldots, \hat{\gamma}_J)^\mathsf{T}$ and $\hat{\boldsymbol{\Gamma}} = (\hat{\Gamma}_1, \ldots, \hat{\Gamma}_J)^\mathsf{T}$ be the vectors of estimated SNP–exposure and SNP–outcome associations, let $\boldsymbol{R}$ be the signed genotype correlation matrix estimated from the 1000 Genomes East Asian reference panel (Data Preprocessing), and let $\boldsymbol{D}_Y = \mathrm{diag}(s_{Y1}, \ldots, s_{YJ})$ contain the standard errors of the SNP–outcome associations. We approximated the sampling covariance of $\hat{\boldsymbol{\Gamma}}$ by $\boldsymbol{\Sigma} = \boldsymbol{D}_Y \boldsymbol{R} \boldsymbol{D}_Y$. The IVW model then becomes

$$\hat{\boldsymbol{\Gamma}} = \beta \hat{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}_J(\mathbf{0}, \phi \boldsymbol{\Sigma}), \tag{3}$$

where ϕ is the residual variance multiplier of a multiplicative random-effects model. The GLS estimate of the causal effect is

$$\hat{\beta}_{\text{GLS-IVW}} = \frac{\hat{\boldsymbol{\gamma}}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \hat{\boldsymbol{\Gamma}}}{\hat{\boldsymbol{\gamma}}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \hat{\boldsymbol{\gamma}}}. \tag{4}$$

Under strong LD, $\boldsymbol{\Sigma}$ can be nearly singular, and inverting it directly is numerically unstable. We therefore truncated the eigendecomposition $\boldsymbol{\Sigma} = \boldsymbol{V} \boldsymbol{\Lambda} \boldsymbol{V}^\mathsf{T}$, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_J$ and orthonormal eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_J$. We retained the set $\mathcal{K}$ of leading eigen-directions that together explain 99.9% of the variance, with a minimum of five directions, and kept only directions whose eigenvalues exceed 10^{-10} times the largest eigenvalue; $k = |\mathcal{K}|$ (formal definition in S1 Text). The truncated matrix $\widetilde{\boldsymbol{\Sigma}} = \sum_{\ell \in \mathcal{K}} \lambda_\ell \boldsymbol{v}_\ell \boldsymbol{v}_\ell^\mathsf{T}$ keeps the k largest eigenvalues of $\boldsymbol{\Sigma}$; by the Eckart–Young theorem, it is the best rank-k approximation of $\boldsymbol{\Sigma}$ in both the Frobenius and spectral norms [CITE: Eckart–Young theorem]. Because $\widetilde{\boldsymbol{\Sigma}}$ is singular when $k < J$, we replaced $\boldsymbol{\Sigma}^{-1}$ by its Moore–Penrose pseudoinverse $\widetilde{\boldsymbol{\Sigma}}^{+} = \sum_{\ell \in \mathcal{K}} \lambda_\ell^{-1} \boldsymbol{v}_\ell \boldsymbol{v}_\ell^\mathsf{T}$, which inverts $\boldsymbol{\Sigma}$ on the retained eigenspace and gives zero weight to the discarded directions. The causal IVW estimate is therefore

$$\hat{\beta}_{\text{GLS-IVW}} = \frac{\hat{\boldsymbol{\gamma}}^\mathsf{T} \widetilde{\boldsymbol{\Sigma}}^{+} \hat{\boldsymbol{\Gamma}}}{\hat{\boldsymbol{\gamma}}^\mathsf{T} \widetilde{\boldsymbol{\Sigma}}^{+} \hat{\boldsymbol{\gamma}}}, \tag{5}$$

and, provided that the denominator is positive, its conditional variance under the working covariance model is

$$\mathrm{Var}\left(\hat{\beta}_{\text{GLS-IVW}} \mid \hat{\boldsymbol{\gamma}}, \boldsymbol{\Sigma}\right) = \frac{\phi}{\hat{\boldsymbol{\gamma}}^\mathsf{T} \widetilde{\boldsymbol{\Sigma}}^{+} \hat{\boldsymbol{\gamma}}}. \tag{6}$$

The causal effect is the slope β, and we tested $H_0 : \beta = 0$. The residual dispersion was estimated as $\hat{\phi} = Q/(J-1)$, where $Q = (\hat{\boldsymbol{\Gamma}} - \hat{\beta}_{\text{GLS-IVW}}\hat{\boldsymbol{\gamma}})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \hat{\beta}_{\text{GLS-IVW}}\hat{\boldsymbol{\gamma}})$ is the residual quadratic form and J is the number of instruments after LD filtering, not the number of retained eigen-directions k. Pleiotropic instruments make the SNP-specific ratio estimates more heterogeneous than sampling error alone would, so the residual variance can exceed that implied by $\boldsymbol{\Sigma}$ (overdispersion). We accommodated this with the multiplicative random-effects model by setting $\widetilde{\phi} = \max(1, \hat{\phi})$, which inflates the standard error under overdispersion but never makes it smaller than under the fixed-effect model ($\phi = 1$) (details in S1 Text). The standard error was $\widehat{\text{SE}}(\hat{\beta}_{\text{GLS-IVW}}) = \{\widetilde{\phi}/(\hat{\boldsymbol{\gamma}}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\gamma}})\}^{1/2}$, and the two-sided P value was obtained by comparing $T = \hat{\beta}_{\text{GLS-IVW}}/\widehat{\text{SE}}(\hat{\beta}_{\text{GLS-IVW}})$ with a t distribution with $J-1$ degrees of freedom.

Causal Egger: GLS MR-Egger with correlated instruments

Causal Egger extends the causal IVW model by adding an intercept [4]. With the association vectors and LD-based covariance matrix defined above, the model is

$$\hat{\boldsymbol{\Gamma}} = \alpha_0 \mathbf{1}_J + \beta\hat{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}_{\text{Egger}}, \qquad \boldsymbol{\varepsilon}_{\text{Egger}} \sim \mathcal{N}_J(\mathbf{0}, \phi_{\text{Egger}}\boldsymbol{\Sigma}), \tag{7}$$

where $\mathbf{1}_J$ is a vector of ones, α_0 is the intercept, and ϕ_{Egger} is the residual variance multiplier. As in causal IVW, the estimated SNP–exposure associations $\hat{\boldsymbol{\gamma}}$ were treated as fixed under the NO Measurement Error (NOME) assumption (S1 Text), and the pseudoinverse $\widetilde{\boldsymbol{\Sigma}}^{+}$ defined above was used as the weighting matrix. Define

$$\boldsymbol{X} = \begin{bmatrix} \mathbf{1}_J & \hat{\boldsymbol{\gamma}} \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{pmatrix} \alpha_0 \\ \beta \end{pmatrix}. \tag{8}$$

Provided that $\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\boldsymbol{X}$ is nonsingular, the GLS estimates are

$$\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}} = \begin{pmatrix} \hat{\alpha}_0 \\ \hat{\beta}_{\text{GLS-Egger}} \end{pmatrix} = \left(\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\boldsymbol{X}\right)^{-1}\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\hat{\boldsymbol{\Gamma}}, \tag{9}$$

with conditional covariance matrix

$$\text{Var}\left(\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}} \mid \hat{\boldsymbol{\gamma}}, \boldsymbol{\Sigma}\right) = \phi_{\text{Egger}}\left(\boldsymbol{X}^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}\boldsymbol{X}\right)^{-1} \tag{10}$$

under the working covariance model. Inference on the slope followed causal IVW. The residual dispersion $\hat{\phi}_{\text{Egger}}$ was the residual quadratic form $Q_{\text{Egger}} = (\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\hat{\boldsymbol{\theta}}_{\text{GLS-Egger}})$ divided by $J-2$; the slope standard error was the square root of the second diagonal element of the covariance matrix, with

ϕ_{Egger} replaced by $\widetilde{\phi}_{\mathrm{Egger}} = \max(1, \hat{\phi}_{\mathrm{Egger}})$; and two-sided P values for $H_0 : \beta = 0$ were obtained from a t distribution with $J - 2$ degrees of freedom. Causal Egger estimates depend on the allele orientation of the instruments, whereas causal IVW estimates do not (Allele orientation).

Causal SMR

Summary-data-based Mendelian randomization (SMR) estimates the relationship between an exposure and an outcome from the summary-level associations of a single genetic instrument with each of them [5]. We applied this approach to chromatin accessibility and gene expression. For each peak–gene pair, we selected the SNP with the smallest nominal caQTL P value among the variants retained after the filtering described above, breaking ties by the largest absolute ratio of the caQTL effect to its standard error. Denoting this SNP by $*$, the effect estimate is

$$\hat{\beta}_{\mathrm{SMR}} = \frac{\hat{\Gamma}_*}{\hat{\gamma}_*}. \tag{11}$$

The SMR test uses a delta-method approximation to the variance of this ratio that accounts for uncertainty in both associations [5]. In our implementation, this approximation omitted the sampling covariance between the caQTL and eQTL effect estimates.

For the selected SNP, we converted the two-sided nominal caQTL and eQTL P values, p_X and p_Y, into nonnegative normal-equivalent statistics:

$$\begin{aligned} z_X &= \Phi^{-1}(1 - \tilde{p}_X/2), \qquad z_Y = \Phi^{-1}(1 - \tilde{p}_Y/2), \\ \tilde{p}_X &= \max(p_X, 10^{-300}), \quad \tilde{p}_Y = \max(p_Y, 10^{-300}), \end{aligned} \tag{12}$$

where Φ is the standard normal cumulative distribution function; numerically, the inverse was evaluated on the upper tail. From these P-derived statistics, we calculated

$$\begin{aligned} \widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{SMR}}) &= |\hat{\beta}_{\mathrm{SMR}}| \sqrt{z_X^{-2} + z_Y^{-2}}, \\ T_{\mathrm{SMR}} &= \frac{z_X^2 z_Y^2}{z_X^2 + z_Y^2}. \end{aligned} \tag{13}$$

The SMR P value was the upper-tail probability of T_{SMR} under an approximate χ^2_1 reference distribution. The sign of the effect was given by the ratio of the aligned association estimates before the allele orientation step (Allele orientation). We did not perform the heterogeneity in dependent instruments (HEIDI) test; SMR significance alone does not distinguish mediation from horizontal pleiotropy or from linkage between distinct causal variants [5].

Causal joint: Cauchy combination

The Cauchy combination test (CCT) is an efficient and powerful technique to combine multiple individual P values, especially when the tests are correlated [6]. Under the null hypothesis, for an individual P value p_k, the transformed statistic follows a standard Cauchy distribution. We calculated the causal joint results by aggregating the three causal methods (causal IVW, causal Egger, and causal SMR) using the Cauchy combination test with equal weights:

$$T = \frac{1}{3}\left\{\tan\left[\pi(0.5 - p_{\mathrm{IVW}})\right] + \tan\left[\pi(0.5 - p_{\mathrm{Egger}})\right] + \tan\left[\pi(0.5 - p_{\mathrm{SMR}})\right]\right\},$$

where p_{IVW}, p_{Egger}, and p_{SMR} denote the nominal P values from GLS-IVW, GLS MR-Egger, and SMR, respectively. The combined P value is

$$p_{\mathrm{joint}} = \frac{1}{2} - \frac{\arctan T}{\pi},$$

computed as $\arctan(1/T)/\pi$ when $T > 0$ for numerical accuracy. For $p < 10^{-15}$, the term $\tan[\pi(0.5 - p)]$ was replaced by its tail approximation $1/(\pi p)$, and for $p > 1 - 10^{-15}$ by $-1/\{\pi(1 - p)\}$. The causal joint results can be understood as the aggregated causal evidence for the studied peak–gene links.

Marginal pseudobulk correlation

We evaluated marginal peak–gene associations using pseudobulk RNA and ATAC profiles within each cell type. The starting point was the CIMA count matrices for all cells (RNA: 6,484,974 cells and 36,326 genes; ATAC: 3,762,242 cells and 338,036 peaks), which were subset to each cell type using the CIMA level-4 annotation. For each modality, individuals were retained if they had at least 100 cells and no more than the median plus three median absolute deviations (MAD scaled by 1.4826) of the per-individual cell number; individuals passing in both modalities were used, yielding 335 individuals for Bn cells, 372 for CD4 T cells, and 346 for CD8 T cells. Counts were summed across cells from the same individual within each cell type to construct gene-by-individual and peak-by-individual pseudobulk matrices (scanpy 1.10.4). For each modality, counts were normalized using the individual-level library size L, the sum of counts across all features of that modality, and transformed as $A = \ln(1 + 10^6 C/L)$, where C is the summed count.

For each candidate peak–gene pair, we calculated Spearman's rank correlation between peak accessibility and gene expression across matched individuals:

$$\widehat{\rho}_{pg}^{\mathrm{PB}} = \mathrm{Cor}_{\mathrm{S}}\left(\boldsymbol{A}_p, \boldsymbol{E}_g\right), \tag{14}$$

where $\boldsymbol{A}_p$ and $\boldsymbol{E}_g$ are the log-transformed CPM vectors for peak p and gene g, respectively, across matched individuals, and $\mathrm{Cor_S}$ denotes Spearman's rank correlation, with tied values assigned average ranks. Two-sided approximate P values were obtained using R's `cor.test` with `method="spearman"` and `exact=FALSE`, which uses a t approximation with $n-2$ degrees of freedom. No covariate residualization was applied at the correlation stage. All pairs in a cell type used the same individuals, with no missing values. P values were adjusted as described in Filtering and multiple testing, and peak–gene pairs with $P_{\mathrm{adj}} < 0.1$ were considered significant.

Marginal multiomics: WNN metacell correlation

We evaluated marginal peak–gene associations using an external PBMC multiome dataset with 80 precomputed weighted nearest-neighbor (WNN) metacell groups. We used the existing WNN metacell groups supplied with the dataset and did not recluster the cells. The same RNA and ATAC metacell profiles were used for all three CIMA cell-type analyses, without restricting metacells to the corresponding cell type. WNN defined the cell groups, whereas peak–gene associations were quantified by correlations across metacells.

TODO: Add the external dataset description and citation, cell QC, the WNN method citation, and details of the original metacell construction and RNA aggregation and normalization; Signac version used for recounting (notes/2_BOARD.md section 3).

To align accessibility features with CIMA, external ATAC fragments were recounted over all 338,036 CIMA ATAC peak intervals in hg38 using Signac's `FeatureMatrix`, after excluding intervals that overlapped other peaks in the input set. ATAC counts were summed within each metacell and normalized to counts per million using the total counts across retained peaks in that metacell. The supplied RNA metacell matrix and group assignments were used as provided. A $\log(1+x)$ transformation was then applied to both the normalized ATAC matrix and the supplied RNA matrix.

For each candidate peak–gene pair, we calculated Spearman's rank correlation between peak accessibility and gene expression across the 80 metacells:

$$\widehat{\rho}_{pg}^{\mathrm{WNN}} = \mathrm{Cor_S}\left(\boldsymbol{A}_p^{\mathrm{MC}}, \boldsymbol{E}_g^{\mathrm{MC}}\right), \tag{15}$$

where $\boldsymbol{A}_p^{\mathrm{MC}}$ and $\boldsymbol{E}_g^{\mathrm{MC}}$ denote the transformed accessibility and expression vectors for peak p and gene g, respectively, across metacells, and $\mathrm{Cor_S}$ denotes Spearman's rank correlation. Metacells served as the units of observation. Two-sided approximate P values were calculated as described in Marginal pseudobulk correlation. No covariate residualization, cell-type adjustment, or weighting by metacell size was applied.

The resulting correlations therefore include variation across PBMC cell types and states. Because one matrix was used for all three cell types, a given pair has the same $\widehat{\rho}^{\mathrm{WNN}}$ and nominal P value in each cell type, but its P_{adj} can differ because the set of pairs tested for each gene differs between cell types. P values were adjusted as described in Filtering and multiple testing, separately for each CIMA cell-type analysis, and peak–gene pairs with $P_{\mathrm{adj}} < 0.1$ were considered significant.

Filtering and multiple testing

After all methods had been applied, we first removed pairs for which any of the five base methods (causal IVW, causal Egger, causal SMR, marginal multiomics and marginal pseudobulk) returned a missing P value. We then removed pairs with $\hat{\phi}_{\mathrm{IVW}} > 10$: at this level of residual dispersion the GLS fit had failed, typically because of outlying instruments or a near-singular LD matrix among highly correlated variants. Causal joint P values were computed for the remaining pairs. Finally, each of the six sets of P values was adjusted by the Benjamini–Hochberg procedure within each gene, separately for each method and cell type [14], giving P_{adj} (Study design and terminology). Discovery used $P_{\mathrm{adj}} < 0.1$.

TODO: Author: one sentence on why BH is applied within each gene (each gene is tested with multiple candidate peaks).

Hypergeometric enrichment

We tested whether the significant peaks of each method were enriched for peaks that overlap existing enhancer and chromatin interaction references: FANTOM5 [CITE: FANTOM5 enhancer atlas] linked enhancers and genes based on enhancer RNA expression; 4DGenome [CITE: 4DGenome] linked enhancers and genes based on physical interactions using chromatin-looping data; EnhancerAtlas [CITE: EnhancerAtlas] reported enhancers using 12 high-throughput experimental methods; cis-eQTLs in whole blood mapped in European-American subjects were obtained from the GTEx Consortium [CITE: GTEx]; promoter capture Hi-C (PCHi-C) captured contacts between gene promoters and distal regions [CITE: PCHi-C in immune cells]; and H3K27ac HiChIP captured contacts among regions marked by an active regulatory histone modification [CITE: H3K27ac HiChIP; Supplementary Table 3 of Nat Genet article 41588_2020_745]. For HiChIP, we used interacting regions with $Q < 0.1$ in the B cell, CD4 T cell and CD8 T cell columns; for PCHi-C, we used the other ends of contacts with CHiCAGO score ≥ 5 in naive B cells, naive CD4 T cells and naive CD8 T cells. For GTEx, we used variants with meta-analysis $P < 0.05$ and removed variants located within transcripts annotated in EnsDb.Hsapiens.v86 (2.99.0). Only the HiChIP and PCHi-C regions were lifted over from hg19 to hg38, using rtracklayer (1.66.0) with the

`hg19ToHg38.over.chain` file; regions that failed to map or mapped to more than one chromosome were discarded, and regions that mapped to multiple segments were represented by the range from the first to the last segment.

TODO: Sources and genome builds of the GTEx, EnhancerAtlas, FANTOM5 and 4DGenome objects (notes/2_BOARD.md section 3).

For each cell type and method, and within each gene, let N be the number of candidate peaks, K the number of these peaks that overlap the gene's reference regions, n_{sig} the number of peaks significant for the method ($P_{\text{adj}} < 0.1$), and k the number of peaks that are both significant and overlapping. Genes with $N > 0$, $K > 0$, $n_{\text{sig}} > 0$ and $N > K$ were included. We summed N, K, n_{sig} and k across genes and performed one global test, coded in R as `phyper(`$\sum k - 1$`, `$\sum K$`, `$\sum N - \sum K$`, `$\sum n_{\text{sig}}$`, lower.tail = FALSE)`, without further multiple-testing correction. The pooled fold enrichment was $(\sum k / \sum n_{\text{sig}})/(\sum K / \sum N)$. Results are given in S1 Table and S2 Table.

Gene ontology analysis

We performed over-representation analysis with the DAVID Functional Annotation Chart [CITE: DAVID]. The input was the union of the genes of MR-only links (Study design and terminology) across the three cell types, 4,509 gene symbols (OFFICIAL_GENE_SYMBOL, *Homo sapiens*, DAVID default background). We used the GOTERM_BP_DIRECT and GOTERM_MF_DIRECT categories; DAVID exports only terms with $P < 0.1$, and we report the DAVID FDR. Of the input genes, 4,031 had a GO Biological Process annotation and 4,185 had a GO Molecular Function annotation. Fig 3B and C show the top 10 terms in each category ranked by DAVID FDR. The analysis was run on 2026-10-19.

TODO: DAVID knowledgebase version.

GRN overlap

We compared the links of each method with the CIMA SCENIC+ (1.0.1) enhancer-driven regulons (eRegulons) [1] [CITE: SCENIC+]. We used two sets: the eRegulons active in the cell type (cell-type eRegulons) and the subset of these that belong to the high-quality eRegulons of the corresponding lineage (high-quality eRegulons). From each set we extracted the unique region–gene pairs. For each cell type and method, we built a 2×2 table of all tested pairs in the cell type, cross-classified by whether the pair was significant ($P_{\text{adj}} < 0.05$) and whether it was present in the eRegulon set, and applied Fisher's exact test. We report the odds ratio (Fig 3E; S8 Fig).

Heritability enrichment analysis from S-LDSC

We used stratified LD score regression (S-LDSC; LDSC v1.0.1) [7] to test whether peaks of MR-only links were enriched for disease heritability. The MR-only peaks were the union of the peaks of MR-only links (Study design and terminology) across the three cell types. The comparison set was all candidate peaks (316,798 peaks in hg19). Peaks were lifted over from hg38 to hg19 with rtracklayer (1.66.0), keeping only one-to-one mappings on chromosomes 1–22, extended by 500 bp on each side and merged. The annotation was the set of SNPs located in the extended peaks. We fitted S-LDSC separately for the MR-only annotation and for the all-candidate-peak annotation, each together with the 97 baselineLD v2.2 annotations, with `--ld-wind-cm 1`, LD scores restricted to HapMap3 SNPs, and `--overlap-annot --print-coefficients`. We used the 1000 Genomes Phase 3 European LD reference data and the standard regression weights that exclude the MHC region; the reference files were obtained from Zenodo (doi:10.5281/zenodo.10515792). Enrichment was defined as the proportion of heritability explained by the annotation divided by the proportion of SNPs in the annotation, and we report the enrichment estimate and its P value for the tested annotation.

We report seven immune-mediated diseases: ankylosing spondylitis (AS), Crohn's disease (CD), inflammatory bowel disease (IBD), rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), multiple sclerosis (MS), and type 1 diabetes (T1D). Summary statistics for IBD [15], RA [16], SLE [17] and T1D [18] were obtained from the collection of 107 independent GWAS summary statistics on Zenodo (`https://zenodo.org/records/10515792/files/sumstats_indep107.tgz`). Summary statistics for AS were obtained from the Pan-UK Biobank (`https://pan.ukbb.broadinstitute.org/`), those for CD from the International IBD Genetics Consortium (`https://www.ibdgenetics.org/`), and those for MS from ImmunoBase [19]. The sample sizes and sources of the seven GWAS are given in S4 Table.

GWAS overlap

To show individual examples, we selected strong MR-only links, defined as links with $P_{\mathrm{adj}} < 0.001$ for at least one of the four causal methods and $P_{\mathrm{adj}} \geq 0.05$ for both marginal methods, whose peak contained at least one SNP with $P < 5 \times 10^{-8}$ in any of the seven GWAS analyzed. In Fig 4B, links are drawn as significant at $P_{\mathrm{adj}} < 0.05$.

Motif enrichment analysis

We performed motif enrichment analysis to determine whether transcription factor binding motifs were overrepresented in significant peaks compared with background peaks. For each cell type (Bn cells, CD4 T

cells, CD8 T cells) and each of the six methods, we used the peaks of links with $P_{\text{adj}} < 0.1$ as the foreground set. The universe was the peaks of the candidate pairs that were also present in the ATAC object of that cell type (316,117 peaks in Bn cells), and the background was 50,000 peaks drawn from this universe with `MatchRegionStats`, matched for GC content and $\log(1 + \text{total accessibility})$. Motif occurrences were annotated with `AddMotifs` in Signac (1.14.0) [CITE: Signac] using the JASPAR2020 CORE vertebrate collection (JASPAR2020 0.99.10; latest version of each matrix, 746 motifs) [CITE: JASPAR2020] and the hg38 genome (BSgenome.Hsapiens.UCSC.hg38 1.4.5). We used `FindMotifs` in Signac, which computes P values with a hypergeometric test, and adjusted the P values by the Benjamini–Hochberg procedure across motifs within each cell type and method. Motifs with adjusted $P < 0.05$ and fold enrichment > 1 were considered enriched. A motif was MR-specific if it was enriched for at least one of causal IVW, causal Egger and causal SMR and for neither marginal method; causal joint was not used in this definition. For Table 1, we selected for each cell type the two MR-specific motifs with the smallest adjusted P among the causal methods, skipping motifs without a clear literature report.

Allele orientation

Before fitting causal IVW and causal Egger, we flipped the sign of $\hat{\gamma}_j$ and $\hat{\Gamma}_j$, together with the corresponding row and column of $\boldsymbol{R}$, for every SNP with $\hat{\gamma}_j < 0$, so that all SNP–exposure associations were positive. This orientation does not change causal IVW estimates but does change causal Egger estimates. We compared causal Egger with and without this orientation across 176,545 pairs (S2 Fig).

Sensitivity analysis: Overlapping samples

Consider the two marginal models of the caQTL and eQTL studies, which provide the exposure and outcome associations in the MR framework. For each SNP separately, the marginal associations can be represented as

$$X_i = \mu_{Xj} + \gamma_j G_{ij} + \varepsilon_{ij}^X,$$
$$Y_i = \mu_{Yj} + \Gamma_j G_{ij} + \varepsilon_{ij}^Y,$$

where μ_{Xj} and μ_{Yj} are intercepts, and γ_j and Γ_j are the population-level SNP–exposure and SNP–outcome associations, respectively. MR study is usually base on Two-Sample data, where the SNP-exposure estimates and SNP-outcome estimates come from two independent but homogeneous samples. In this ways, the residuals ε_{ij}^X and ε_{ij}^Y are independent. However, in our study, caQTL and eQTL come from the same population, leading to some potential bias. In order to measures of the bias, we do the following sensitivity test for each of cell types separately.

We first performed LD pruning (`--indep-pairwise 500kb 50 0.1`, MAF ≥ 0.01, with MAF computed in the reference panel) on the caQTL variants on chromosomes 1–22 that matched the EAS reference panel. We further selected SNPs under the null (caQTL nominal $P > 0.05$ and eQTL nominal $P > 0.05$). The marginal associations are under the null, there is no evidence to prove the marginal associations (γ_j and Γ_j) are nonzero values. Therefore, we can use the correlation of Z scores to approximate the correlation of residuals:

$$z_{Xj} = \frac{\hat{\gamma}_j}{s_{Xj}}, \qquad z_{Yj} = \frac{\hat{\Gamma}_j}{s_{Yj}}.$$

We calculated the genome-wide correlation of z_{Xj} and z_{Yj} for each cell type by pooling all (pair, SNP) observations: 17,074,938 (Bn cells), 33,263,695 (CD4 T cells) and 21,537,822 (CD8 T cells) observations. We also calculated the correlation of z_{Xj} and z_{Yj} for each peak–gene pair with more than two SNPs after LD pruning (547,452, 1,051,953 and 686,829 pairs, respectively), tested with a t test with $n - 2$ degrees of freedom. In this analysis, alleles were aligned by flipping the sign of the eQTL slope for variants with reversed alleles, rather than by removing them as in the main MR analysis. The distributions of correlations and P values are shown in S1 Fig.

Acknowledgments

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Supporting information

All supporting figures and tables are provided in a separate file (`supplementary.pdf`).

S1 Text. Supplementary Methods. MR model and summary statistics, IVW and MR-Egger for uncorrelated variants, and the truncated eigendecomposition used by the GLS estimators.

S1 Fig. Null-SNP diagnostics in Bn, CD4 T cells, and CD8 T cells.

S2 Fig. MR-Egger estimates with and without allele orientation in Bn, CD4 T cells, and CD8 T cells.

S3 Fig. Overlap and effect directions of significant links in Bn and CD8 T cells.

S4 Fig. Sharing of significant links among Bn, CD4 T cells, and CD8 T cells for each method.

S5 Fig. Overlap of MR-only links with regulatory references compared with all candidate peaks.

S6.1 Fig. Pairwise relationships among the five methods with an effect estimate in Bn.

S6.2 Fig. Pairwise relationships among the five methods with an effect estimate in CD4 T cells.

S6.3 Fig. Pairwise relationships among the five methods with an effect estimate in CD8 T cells.

S7.1 Fig. Genome-wide association signals of the six methods in Bn.

S7.2 Fig. Genome-wide association signals of the six methods in CD4 T cells.

S7.3 Fig. Genome-wide association signals of the six methods in CD8 T cells.

S8 Fig. Overlap of significant links with the CIMA regulatory network in Bn and CD8 T cells.

S1 Table. Pooled hypergeometric enrichment p-values across regulatory references.

S2 Table. Pooled regulatory-reference fold enrichment across cell types and methods.

S3 Table. Data overview for the three cell types analyzed.

S4 Table. GWAS used in the S-LDSC analysis and in the GWAS-overlap tracks.